Introduction to Programming
CSL-100/ MAL-400

Full Marks 100 Time 3 hours

- Read all question carefully
- Answer on the space given in the question paper itself
- Answer all questions
- The questions are self explanatory do not ask question to invigilators
- Return the question paper on completion

Q1)

a. Consider the following C function definition:

```
int Trial (int a, int b, int c)
{
if ((a > = b) && (c < b)) return b;
else if (a > = b) return Trial (a,c,b);
else return Trial (b,a,c);
}
```

The function Trial is used to find :

(a) Finds the maximum of a, b, and c

(b) Finds the minimum of a, b and c

(c) Finds the middle number of a, b, c

(d) None of the above

b. Consider the following function written in the C programming language.

```
void foo (char *a)
{
while(*a  &&  *a != ' ')
{
putchar (*a);
a=a+1;
}
}
int main(){
```

```
foo("abcd");

}
```

The output of the above function on input “ABCD EFGH” is_____

C. In the following code in which order the function would be called?

```
a = f1(23,14) * f2(12/4) + f3();
```

D. What is the output of this program?

```
#include <stdio.h>
int main(){
   FILE *fp;
   char *str;
   fp=fopen("demo.txt","r");// demo.txt :you are a good programmer
   while(fgets(str,6,fp)!=NULL)
   puts(str);
   fclose(fp);
   return 0;
}
```

A. you are a good programmer
B. e a good programmer
C. you ar
D. you are

Q2 Fill in the blanks:

a.
```
#include <stdio.h>

int main() {
   int x = 10;
   int *ptr;

   // 1. Declare a pointer variable named 'ptr' and initialize it with the variable 'x'.
   __________ = &x;

   // 2. Use the pointer 'ptr' to change the value of 'x' to 20.
   ______ = __________;

   // 3. Print the value of 'x' using the pointer 'ptr'.
```

```
    printf("The value of x is: %d\n", _________);

    // 4. Declare a new integer variable 'y' and assign a value of 30.
    int y = ________;

    // 5. Declare a pointer 'ptr_y' and assign the 'y' to it.
    int *ptr_y = &_____;

    // 6. Print the value of 'y' using the pointer 'ptr_y'.
    printf("The value of y is: %d\n", __________);

    // 7. Allocate memory for an integer using the 'malloc' function and assign its address
to 'ptr'.
    ptr = ( _____ ) malloc ( __________ );

    // 8. Use the pointer 'ptr' to store the value 42 in the dynamically allocated memory.
    _______ = _________;

    // 9. Print the value stored in the dynamically allocated memory using the pointer 'ptr'.
    printf("The value in dynamically allocated memory is: %d\n", _________);

    // 10. Free the dynamically allocated memory.
    _________( _______ );

    return 0;
}
```

b.
```
#include <stdio.h>

int main() {
int numbers[] = {1, 2, 3, 4, 5};
int *ptr,*ptr_matrix;

  // 1. Declare a pointer 'ptr' and initialize it with the first element of the 'numbers' array.
    _________ = _____________;

    // 2. Use the pointer 'ptr' to print the value of the second element of the 'numbers'
array.
    printf("The value of the second element is: %d\n", _________);

    // 3. Increment the pointer 'ptr' to point to the next element in the 'numbers' array.
    _________++;

    // 4. Print the value that 'ptr' is now pointing to.
```

```
    printf("The value pointed to by ptr is: %d\n", _________);

    // 5. Declare a new integer array 'matrix' with 2 rows and 3 columns.
    int matrix[2][3] = {{10, 20, 30}, {40, 50, 60}};

    // 6. Declare a pointer 'ptr_matrix' and initialize it with the first row in the 'matrix' array.
    _________ = ___________;

    // 7. Print the value of the element in the second row and third column using
'ptr_matrix'.
    printf("The value in the second row, third column is: %d\n", _________);



    // 8. Print the value of the element in the second row and third column using
'ptr_matrix'.
    printf("The value in the second row, second column is: %d\n", _________);

    return 0;
}
```

Q3. Given an array Arr[] of N integers. Find the contiguous sub-array(containing at least one number) which has the maximum sum and return its sum.

Input:

N = 5

Arr[] = {1,2,3,-2,5}

Output:

9

Explanation:

Max subarray sum is 9 of elements (1, 2, 3, -2, 5) which is a contiguous subarray.

Input:

N = 4

Arr[] = {-1,-2,-3,-4}

Output:

-1

Explanation:

Max subarray sum is -1 of element (-1)

Q3) Write a program to find out if two strings are anagrams or not.

Note : anagrams:- Two strings are called anagrams to each other if they have the same characters with equal occurrences in them but can have different orders of characters.

Input:

Enter first word: Smartest

Enter second word : Mattress

Output:

The words are anagrams

Q4) This question involves creating a function (isMagicSquare) to check if a given square matrix is a magic square. A magic square is a square matrix in which the sums of the numbers in each row, each column, and both main diagonals are the same. The main function dynamically allocates memory for the square matrix, inputs values, calls the functions, prints the result, and then frees the dynamically allocated memory.